



Promoting effect of cyano groups attached on g-C₃N₄ nanosheets towards molecular oxygen activation for visible light-driven aerobic coupling of amines to imines

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ABSTRACT

Molecular oxygen activation shows a great potential for a wide range of applications in organic transformations, but remains a challenge owing to the high inertness from its high energy electrons. In this work, we firstly report the dramatically promoting effect of cyano groups on visible light-driven molecular oxygen activation over cyano-decorated g-C₃N₄ nanosheets (DCNNS) fabricated by a facile and green thermal post-treatment approach from g-C₃N₄ nanosheets (CNNS). Experimental studies and theoretical calculation reveal that the homojunction-like structure is constructed in DCNNS owing to the introduction of cyano-groups, which can facilitate the generation of hot carriers, and thus promotes the O₂^{•−} production. The optimized DCNNS exhibits 2.4 times higher activity than pristine CNNS for the visible light-driven oxidative coupling of benzylamine to imine, an industrially important transformation as a probe reaction. The discovered cyano groups-promoting visible light-driven molecular oxygen activation strategy has been extended to coupling of various amines to their corresponding imines. The present work could open a new horizon for designing highly efficient g-C₃N₄-based catalysts for diverse organic transformations via molecular oxygen activation.

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1. Introduction

Nowadays, with the growing awareness of exploiting renewable energies, the development of sunlight-driven reactions for organic synthesis, which is considered as a promising strategy to replace some thermal-driven chemical processes, has drawn increasing attention [1]. For a light-driven reaction system, the photocatalyst is of key importance for triggering the following chemical process through creating hot carriers on its surface. From references, the metallic oxidations [2,3], metallic sulfides [4], plasmonic metal materials [5], had been demonstrated to be efficient photocatalysts for visible light-driven reactions. Despite these achievements significantly push forward with the development of photocatalytic organic synthesis, the intrinsic drawbacks, such as using precious metals as active sites and relatively low activity, seriously limit their large-scale applications [6–9]. For these reasons, developing an efficient, environment-friendly and sustainable heterogeneous

photocatalyst is a quite significant but challenging task for academicians and industrialists [10–13].

Among all the photocatalysts reported by now, graphitic carbon nitride (g-C₃N₄) represents one of the most promising candidate for its natural abundance, chemical and thermal stability, non-toxicity, facile preparation and metal-free nature and has been widely applied in almost all kinds of photocatalysis branches, including but not limited to water splitting [14–17], solar fuel production [18,19], contaminant elimination [20,21] and important organic reactions [10,22,23]. A variety of conventional organic reactions, such as ketalization reaction [24], aminoformylation of nitroarenes [25], hydrogen transfer reaction [26] and oxidative coupling [27–30], have been realized under ambient temperature conditions using g-C₃N₄-based materials as photocatalysts. For light-driven oxidative-coupling process and the other aerobic reactions, the photocatalytic performance highly depends on the activation of oxygen (O₂) molecule, which is the vital step in this kind of transformation processes. In this context, various strategies have been proposed to activate O₂ molecule efficiently, including constructing heterojunction [31–33], order-disorder interfaces

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engineering [28], oxygen-containing group modification [30], and so on. However, to the best of our knowledge, the introduction of cyano ($-\text{C}\equiv\text{N}$) groups on $\text{g-C}_3\text{N}_4$ nanosheets for boosting O_2 activation has not been reported yet.

Inspired by previous works concerning the promoting effect of carbonyl ($-\text{C}=\text{O}$) groups in $\text{g-C}_3\text{N}_4$ framework on O_2 activation [28–30] and the similar properties of $-\text{C}\equiv\text{N}$ groups with $-\text{C}=\text{O}$ groups in some respects, we deduced that the implantation of $-\text{C}\equiv\text{N}$ groups in $\text{g-C}_3\text{N}_4$ framework would also promote O_2 activation. Bearing these in mind, we concentrate our attention on the development of new approaches to anchoring $-\text{C}\equiv\text{N}$ groups on the framework of $\text{g-C}_3\text{N}_4$ nanosheets (CNNS) and the catalytic mechanism of as-synthesized materials for O_2 activation.

Herein, we have successfully prepared homojunction-like cyano groups-decorated $\text{g-C}_3\text{N}_4$ nanosheets (DCNNS) through a facile and green thermal post-treatment method from $\text{g-C}_3\text{N}_4$ nanosheets (CNNS) and the as-synthesized DCNNS photocatalyst exhibits remarkably enhanced performance for visible light-driven O_2 activation with respect to original CNNS. Although there are many researches concerning the introduction of $-\text{C}\equiv\text{N}$ groups into $\text{g-C}_3\text{N}_4$ framework [34–37], as far as we concerned, there is only a few report about the anchoring of $-\text{C}\equiv\text{N}$ groups on $\text{g-C}_3\text{N}_4$ nanosheets without the assistance of any other agent [38]. Further characterizations and density functional theory (DFT) calculations results demonstrate that the implantation of $-\text{C}\equiv\text{N}$ groups can alter band structure of partial CNNS, and thus, form a homojunction-like catalytic material. In this material, hot excitons, as shackles for hot-carrier generation, can dissociate readily into free charges originated from the $-\text{C}\equiv\text{N}$ groups serve as an electron acceptor. The optimized photocatalyst exhibits 2.4 times higher activity than original CNNS in oxidative coupling of benzylamine to imine, a probe reaction. The cyano groups promoting strategy can be extended to various amines to their corresponding imines, one kind of the most important intermediates for fine chemicals. This work also opens a new horizon for designing highly efficient metal-free catalysts with versatile applications via molecular oxygen activation.

2. Experimental section

2.1. Materials and reagents

Dicyandiamide (DCDA) and acetonitrile were purchased from Tianjin Guangfu Fine Chemical Research Institute, China. NH_4Cl was supplied by Tianjin Bodi Chemical Engineering Co., Ltd., China. Benzylamine was obtained from Sinopharm Chemical Reagent Co., Ltd. 5,5-dimethyl-1-pyrroline N-oxide (DMPO), benzonitrile, 4-methoxybenzylamine, 4-fluorobenzylamine, 2-methoxybenzylamine, 4-chlorobenzylamine and 4-bromobenzylamine were provided by Aladdin. 4-methylbenzylamine, 3-methylbenzylamine and 2,4-dichlorobenzylamine were obtained from Macklin. Terpineol was supplied by Tianjin Damao Chemical Reagent Factory. Ultrapure O_2 gas (purity 99.999%) was purchased from Dalian Guangming Special Gas Co., Ltd., China. All reagents used in this study were at least in analytical grade without further purification. De-ionized water was used for all experiments.

2.2. Preparation of photocatalysts

$\text{g-C}_3\text{N}_4$ nanosheets (CNNS) was prepared according to a previously reported procedure [1], with slight modifications. In detail, 2 g DCDA and 10 g NH_4Cl were added into 160 mL de-ionized water, and the mixture was placed into a sonic bath for 5 min to obtain a homogeneous solution. The solution was kept in a drying

oven to evaporate the water at 105 °C. To increase the surface area of CNNS, the obtained dried mixture was put in a crucible with a semi-open cover and heated at 550 °C for 4 h with a rate of 3 °C/min in a muffle furnace. Cyano-groups decorated $\text{g-C}_3\text{N}_4$ nanosheets (DCNNS) was prepared by thermally treating CNNS in N_2 atmosphere. In detail, 0.2 g as-prepared CNNS was heated in a nitrogen atmosphere at different temperature of 450, 500, 550 and 600 °C for 4 h with a rate of 20 °C/min. According to the different heating temperature, the final samples were denoted as DCNNS-450, DCNNS-500, DCNNS-550 and DCNNS-600. For comparison, bulk $\text{g-C}_3\text{N}_4$ (BCN) was synthesized by directly heating DCDA (2 g) in a crucible with a semi-open cover at 550 °C for 4 h with a rate of 3 °C/min in a muffle furnace. Bulk cyano-groups decorated $\text{g-C}_3\text{N}_4$ (BDCN-550) was prepared as follows: 0.2 g as-prepared bulk $\text{g-C}_3\text{N}_4$ was heated in a nitrogen atmosphere at 550 °C for 4 h with a rate of 20 °C/min.

2.3. Photocatalyst characterizations

Transmission electron microscopy (TEM) images were obtained by using a Tecnai F30 TEM instrument (FEI). X-ray powder diffraction (XRD) patterns were taken on a Rigaku Corporation SmartLab 9 X-ray diffractometer equipment using $\text{Cu K}\alpha$ source ($\lambda = 1.5406 \text{ \AA}$). Fourier transform infrared (FTIR) spectroscopy characterization of catalysts was performed on using a Bruker EQUINOX55 infrared spectrometer. X-ray photoelectron spectroscopy (XPS) measurements were obtained using an ESCALAB 250 XPS system with a monochromatized $\text{Al K}\alpha$ X-ray source (15 kV, 150 W, 500 mm, pass energy = 50 eV). All binding energies were calibrated to the C 1s peak at 284.6 eV of the surface adventitious carbon. Nitrogen adsorption and desorption isotherms were measured on a Beishide apparatus of model 3H-2000PS1 system at 77 K. Before the measurement, all samples were degassed at 130 °C for 8 h. The specific surface areas were calculated via the BET method, and the pore size distributions were calculated from an adsorption branch of the isotherm via the BJH model. The UV–Vis diffuse reflectance spectra (DRS) were recorded on a UV–Vis spectrometer (JASCO V-550) and were converted from reflection to absorbance by the Kubelka–Munk method. The room-temperature photoluminescence (PL) spectra were investigated using a fluorescence spectrometer (Hitachi F-7000) equipped with a 150 W Xe lamp with an excitation wavelength of 370 nm. The C/N atomic ratio was determined using an elemental analyzer (Vario EL, Germany) with speedy quantitative analysis of CHNS. The room-temperature electron spin resonance (ESR) spectra were collected with a Bruker A200 spectrometer. A microwave bridge frequency of 9.5 GHz and 100.0 kHz modulation frequency.

2.4. Photoelectrochemical measurements

The photoelectrochemical investigations were performed on a CHI660E workstation (Chenhua, China) with a conversional three-electrode cell. In this electrochemical system, the sample coated indium tin oxide (ITO) glass, a Pt pole, an Ag/AgCl electrode (saturated KCl solution) and 0.5 mol L^{-1} Na_2SO_4 aqueous solution were used as the working electrode, the counter-electrode, the reference electrode, and the electrolyte, respectively. For fabricating a working electrode, an ITO glass ($2.0 \times 2.0 \text{ cm}^2$) was coated by as-synthesized sample slurry, which had been prepared by mixing 10 mg photocatalyst with 500 μL terpineol with the assistance of sonication. Specifically, the slurry was dipped (50 μL) and spin-coated (2000 r min^{-1} , 30 s) onto an ITO glass followed by the drying at 100 °C for the next coating. The whole coating process was repeated five times.

2.5. Spin trapping-ESR detection of superoxide radical anion $O_2^{\cdot-}$

DMPO was used as the trapping agent to detect $O_2^{\cdot-}$. In detail, 5 mg of photocatalyst was dispersed in 2.5 mL of methanol with the assistance of ultrasound. The mixture was bubbled with oxygen for 0.5 h followed by the addition of 14 μ L of DMPO. Finally, the mixture was irradiated under a Xe lamp for 0.5 min, and then was analyzed by ESR.

2.6. Density functional theory (DFT) calculations

The electron band structures of CNNS and DCNNS were carried out using Dmol3 module in Material Studio 6.5 software package. A two dimensional (2D) melon sheet was constructed to represent the incomplete polycondensation of melon, with a large vacuum space of 15 Å to separate two neighboring 2D melon sheets. A cubic cell was adopted for the carbon nitrides materials with experimental lattice constants of $a = 16.7$ Å and $b = 12.4$ Å. The generalized gradient approximation (GGA) with Perdew-Burke-Ernzerhof (PBE) function was employed for the DFT exchange correlation energy, a 340 eV cutoff was assigned to the plane-wave basis set, and the Brillouin zone was sampled by $3 \times 3 \times 4$ -points.

2.7. Photocatalytic measurements for oxidative couplings of amines

The photocatalytic oxidative coupling of amines were carried out in acetonitrile (CH_3CN) solution in O_2 atmosphere (1 atm) under visible light irradiation ($\lambda > 420$ nm). Typically, 10 mg photocatalyst, 0.1 mmol benzotrifluoride and 0.1 mmol substrate were dispersed into 5 mL CH_3CN solution in a 10 mL round-bottom flask. Prior to illumination, the reaction system was saturated with ultrapure O_2 gas, followed by illumination with a 300 W Xe lamp (PLS-SXE300/300UV, PerfectLight, Beijing) equipped with a 420 nm cut-off filter. The temperature of the reaction solution was carefully maintained at 25 °C during the entire experiment with a homemade thermostatic control device. After the irradiation, the photocatalyst was removed by centrifugation, the remaining liquid was examined by GC and GC-MS. The conversion and selectivity were determined using benzotrifluoride as internal standard.

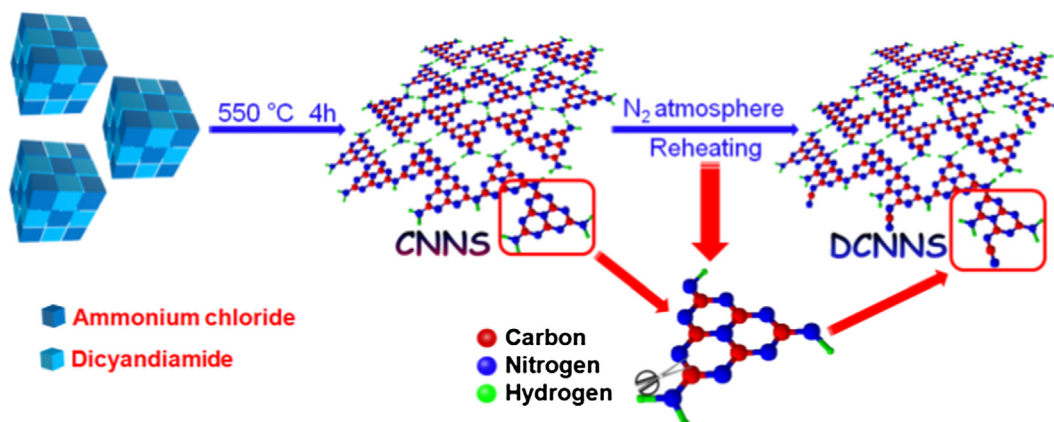
3. Results and discussion

3.1. Preparation and characterization

The plausible fabrication procedures of DCNNS are schematically depicted in Scheme 1. In brief, the CNNS was primarily prepared, according to previous reports with slight modification

[39], through thermally polymerizing the mixed crystals of NH_4Cl and dicyandiamide in static air condition. Then, the CNNS was re-heated at different target temperature under inert atmosphere. After cooling naturally, the DCNNS was successfully synthesized. The as-obtained materials are designated as DCNNS- x , in which x denotes the corresponding post-treating temperature (°C). The morphology of DCNNS-550 was analyzed by TEM. From Fig. 1a, DCNNS-550 features a crinkly nanosheet structure with a size of several micrometers. To examine the crystal structure of CNNS and DCNNS, X-ray diffraction (XRD) characterizations were conducted, all samples display two typical diffraction peaks, which locate at 13.0° and 27.3° , attributed to in-planar structural arrangements of heptazine units and interlayer stacking motif, respectively (Fig. 1b) [14]. Compared to the parent CNNS, the two peaks of DCNNS gradually became weaker with the treatment temperature increases, indicating the loss of ordered structures within the framework. Moreover, the chemical structures of all the samples were analyzed by Fourier transform infrared (FT-IR) spectroscopy (Fig. 1c). The pristine CNNS features adsorption peaks located at 810, 900–1800 and 3000–3500 cm^{-1} , ascribed to the out-of-plane breathing mode of heptazine rings, skeletal stretching modes of $N-C=N$ heterocycles and the vibrational modes of $N-H$ bonds, respectively [18]. For the DCNNS, a new absorption band at 2178 cm^{-1} related to the asymmetric stretching vibration of $-C\equiv N$ groups [34,35], indicating the presence of $-C\equiv N$ groups. Interestingly, the peak intensity of the $-C\equiv N$ group gradually increases along with the increasing treatment temperature.

X-ray photoelectron spectroscopy (XPS) measurements (Supporting Information, Fig. S1) were applied to determine the surface chemical states, and the analysis results are listed in Table S1 and S2. In the C 1s spectrum, three peaks centering at 284.6, 286.2 and 288.1 eV are assigned to standard reference carbon, carbon atom attached to $-NH_x$ on the edges of heptazine units and sp^2 -hybridized carbon in $N-C=N$, respectively. Notably, the 286.2 eV peak of DCNNS-550 shows increased intensity in comparison with that of CNNS, further confirming the formation of $-C\equiv N$ groups because $-C\equiv N$ groups possess similar binding energies to $C-NH_x$ in C 1s spectrum [34]. Correspondingly, the peak-area percentage of 286.2 eV peak increases from 3.30% to 7.48%, while the peak-area percentage of 288.1 eV peak decreases from 63.67% to 59.04%, clearly confirming that the generation of cyano groups is most likely due to the breakage of $N-C=N$ heterocycles during thermal treatment process [35]. Elements analysis (EA) was conducted to investigate the elemental compositions. As listed in Table S3, atomic ratio C/N for DCNNS-550 is 0.661, slightly higher than that of CNNS (0.648), and this weak increase could be ascribed to that the removal of nitrogen atoms is slightly higher than that of



Scheme 1. The fabrication process of DCNNS.

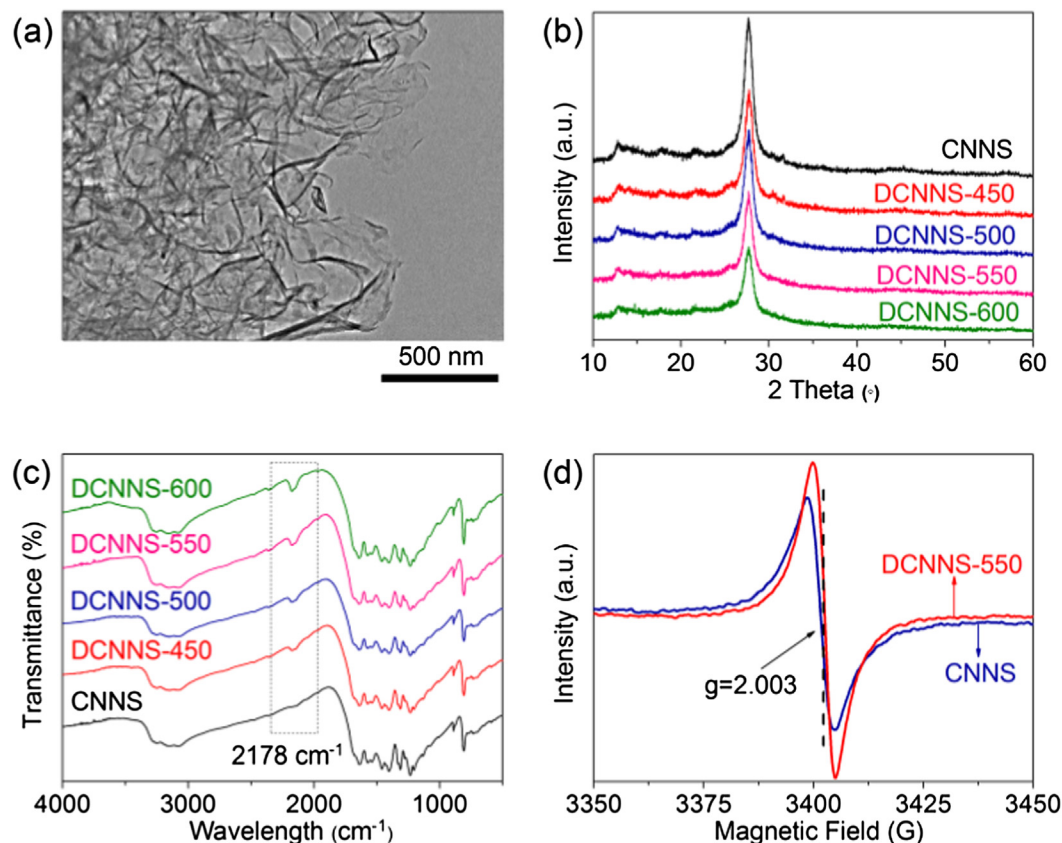


Fig. 1. (a) TEM image of DCNNS-550. (b) XRD patterns and (c) FT-IR spectra of CNNS, DCNNS-450, DCNNS-500, DCNNS-550 and DCNNS-600. (d) Room-temperature ESR spectra of CNNS and DCNNS-550 without the light irradiation.

carbon atoms during thermal treatment. Additionally, $\text{—C}\equiv\text{N}$ groups decorated bulk $\text{g-C}_3\text{N}_4$ (DBCN-550) was prepared by thermally treating bulk $\text{g-C}_3\text{N}_4$ (BCN) at 550°C in N_2 atmosphere, and evidenced with XRD, FT-IR, XPS and EA (Figs. S2–S4 and Tables S3–S5). Room-temperature electron spin resonance (ESR) was also performed to detect the spin state of unpaired electrons (Fig. 1d), both of CNNS and DCNNS-550 exhibit a single Lorentzian line with similar g values of 2.003, which can be ascribed to the unpaired electron on sp^2 -carbon atoms within the π -bonded aromatic rings [19]. The DCNNS-550 shows notably enhanced peak intensity compared with CNNS in the dark, which could be contributed to redistribution of π -electrons caused by the formation of $\text{—C}\equiv\text{N}$ groups.

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.jcat.2018.08.018>.

From Refs. [28–30], we deduced that the implantation of $\text{—C}\equiv\text{N}$ groups in $\text{g-C}_3\text{N}_4$ framework would also promote O_2 activation. To confirm the conjecture, we employed 5,5-dimethyl-1-pyrroline N-oxide (DMPO) to examine the O_2 activation properties at first. From Fig. 2a, the distinct characteristic signals of DMPO- $\text{O}_2^{\cdot-}$ adducts, a nearly 1:1:1:1 quartet pattern, for CNNS and DCNNS-550 were observed, suggesting the generation of $\text{O}_2^{\cdot-}$ on the surface of two samples under visible light irradiation [3]. Moreover, the $\text{O}_2^{\cdot-}$ signal intensity of DCNNS-550 is prominently stronger than that of CNNS, clearly demonstrating that the introduction of $\text{—C}\equiv\text{N}$ groups can significantly promote the activation of oxygen molecule. Fig. 2b displays the steady-state photoluminescence (PL) spectra of different samples performed at room temperature. The steady-state PL intensities of DCNNS apparently lower than that of CNNS, indicating their improved separation efficiency of photoexcited charge carriers [16]. In addition, it can be found that the PL intensity of DCNNS decreases gradually and the peak shifts

to long wavelength with the treatment temperature increasing. To evaluate the interfacial charge transfer capability, photocurrent response of the sample loaded on indium tin oxide (ITO) glass was performed. The photocurrent density of DCNNS-550 is remarkably larger than that of CNNS (Fig. 2c), further suggesting the enhanced charge separation on the DCNNS [40]. Taken together, conclusion could be obtained that the $\text{—C}\equiv\text{N}$ groups in $\text{g-C}_3\text{N}_4$ framework could alter the band structure, which is favorable for accelerating photoexcitation dissociation and facilitating the charge carriers separation.

Thus, to further investigate the effect of $\text{—C}\equiv\text{N}$ groups on the electronic band structure, UV-vis diffuse reflectance spectra (DRS), XPS valence band (VB) spectra were employed and the results are shown in Figs. S5 and S6. Compared with CNNS, the adsorption edge of DCNNS shows progressive red-shift with the increasing of treatment temperature, suggesting the improved light-harvesting capability [35]. Consistent with the UV-vis DRS results, the color of the samples suffers a variation from yellow to brown (Fig. S7). As shown in Fig. S5b, the calculated bandgaps for CNNS and DCNNS determined from the transformed Kubelka-Munk function progressively narrow from 2.61 to 2.06 eV. Based on the facts that only a small number of $\text{—C}\equiv\text{N}$ groups have been implanted and DCNNS still possess the same crystal texture as CNNS, we safely conclude that the introduction of $\text{—C}\equiv\text{N}$ groups partially alter the band structure, thus, a homojunction-like structure is constructed in DCNNS. Combining the XPS VB spectra and calculated bandgaps results, we can obtain the electronic band structures of CNNS and DCNNS-550 (Fig. 2d). The conduction band of the $\text{—C}\equiv\text{N}$ groups decorated part in DCNNS-550 is -0.47 eV, which is upward (more positive potential) by 0.14 eV with respect to the reduction level for free $\text{O}_2^{\cdot-}$ radical ($E^0 = -0.33$ eV vs NHE),

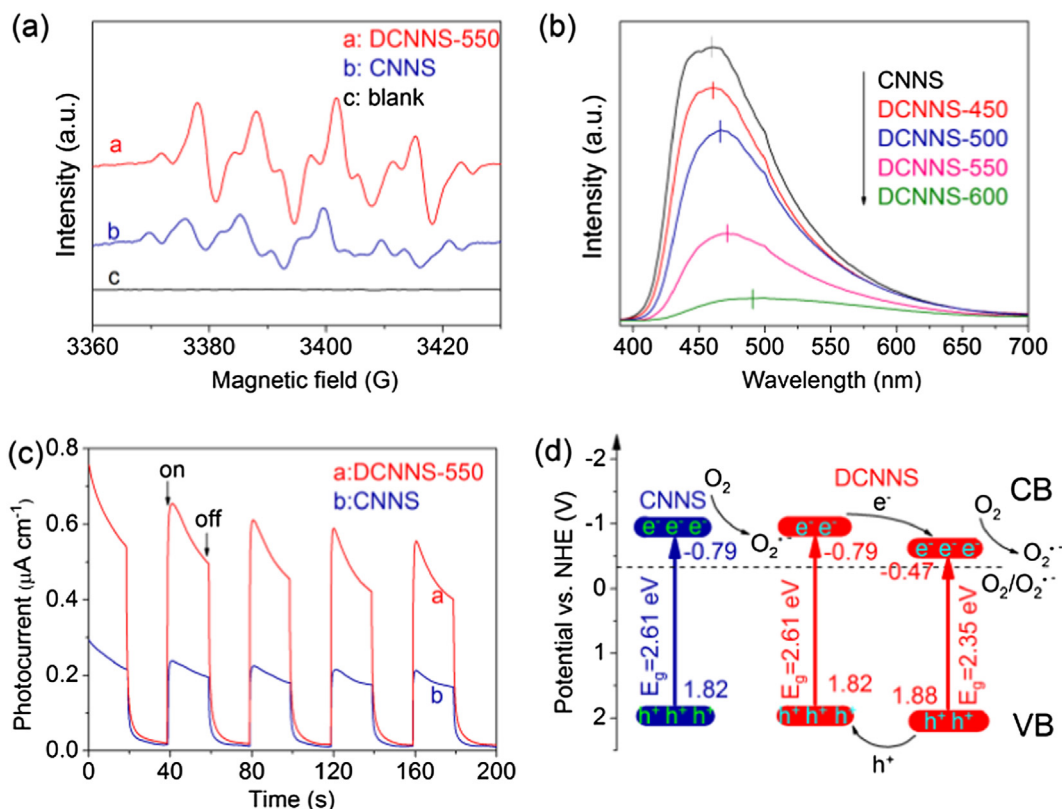


Fig. 2. (a) ESR signals of DMPO- $\text{O}_2^{\bullet-}$ adducts in methanol system for CNNS and DCNNS-550 after 0.5 min irradiation. (b) PL spectra of various samples. (c) Photocurrent response and (d) Electronic band structures of CNNS and DCNNS-550.

and downward (more positive potential) by 0.32 eV compared with that of pristine CNNS. In DCNNS, the lower CB and VB energy levels can facilitate electron migration to $-\text{C}\equiv\text{N}$ groups decorated part and hole injection toward original part, resulting in the enhanced generation of $\text{O}_2^{\bullet-}$ radicals.

3.2. Density functional theory (DFT) calculations

The effect of introduction of $-\text{C}\equiv\text{N}$ groups on electron band structures of CNNS and DCNNS was also investigated through DFT calculation by using Dmol3 module in Material Studio 6.5 software package. The theoretical results are displayed in Fig. 3. The calculated band gap value of DCNNS is 2.26 eV, which is smaller than that of CNNS (2.58 eV). These data are close to the experimental values with slight deviation. The density of states (DOS) reveals that N 2p, C 2s and C 2p orbitals contribute to their CBs, while their VBs consist of N 2s and N 2p orbitals.

3.3. Photocatalytic measurement for probe reaction

The performance of the as-obtained DCNNS catalysts performance for visible light-driven molecular O_2 activation was investigated by a probe reaction concerning the selective oxidative coupling of benzylamine to corresponding imine, which is of industrial importance in a range of chemicals manufacturing process due to its versatile applications [41]. From Table 1, the bare CNNS exhibits relatively low photocatalytic activity with a benzylamine conversion of 26% (Table 1, entry 1). When CNNS was decorated with $-\text{C}\equiv\text{N}$ groups, apparent enhancement of the benzylamine conversion was observed. With the optimal treatment temperature of 550 °C, DCNNS-550 shows the highest benzylamine conversion of 53% (Table 1, entry 4), exceeding that of bare CNNS by 104%. Regarding of the fact that DCNNS-550 possesses the

similar specific surface area and pore size distribution to CNNS (Fig. S8), we can safely conclude that the construction of homojunction-like structure play a highly important role in boosting the photocatalytic activity of DCNNS. Moreover, DBCN-550 (Table 1, entry 7) exhibits 4.3 times higher activity than BCN (Table 1, entry 6), while it only shows 1.7 times higher specific surface area. These data clearly demonstrate that $-\text{C}\equiv\text{N}$ groups is the dominating factor for boosting the photocatalytic activity in this work. When replacing oxygen with nitrogen, the conversion of benzylamine shows a significant reduction (Table 1, entry 8), suggesting that molecular O_2 is of great importance for oxidative coupling. However, no oxidative coupling of benzylamine takes place either in the absence of light irradiation (Table 1, entry 9) or in the absence of catalyst (Table 1, entry 10). To investigate the reaction kinetics, time-dependent catalytic performances of CNNS and DCNNS-550 for the photocatalyzed oxidation of benzylamines were assessed before the reaction reaches a plateau (Fig. S9a). It can be found that the reaction catalyzed by DCNNS-550 rapidly reaches a conversion of 53% in the first 2 h and the slowly increases up to 8 h. Further, the kinetic rates of CNNS and DCNNS-550 derived from time-dependent catalytic performances are plotted in Fig. S9b, which reveals that the oxidative coupling of benzylamine follows the pseudo-first-order reaction. Notably, the calculated kinetic rate constant of DCNNS-550 reaches 0.268 h^{-1} , and this value is 2.4 times higher than that of CNNS. Moreover, it can be safely said that the enhanced molecular oxygen activation strategy can be definitely extended diverse transformations that taking place via a visible light-driven molecular oxygen activation process. From Ref. [30], the introduction of carbonyl groups into g- C_3N_4 efficiently promoted the oxidative coupling of benzylamine to the corresponding imine (76% conversion for 2 h and 91% conversion for 8 h). In comparison with the reported m- $\text{O}=\text{C}_3\text{N}_4$ -2.3, the developed cyano groups modified g- C_3N_4 catalyst in this work

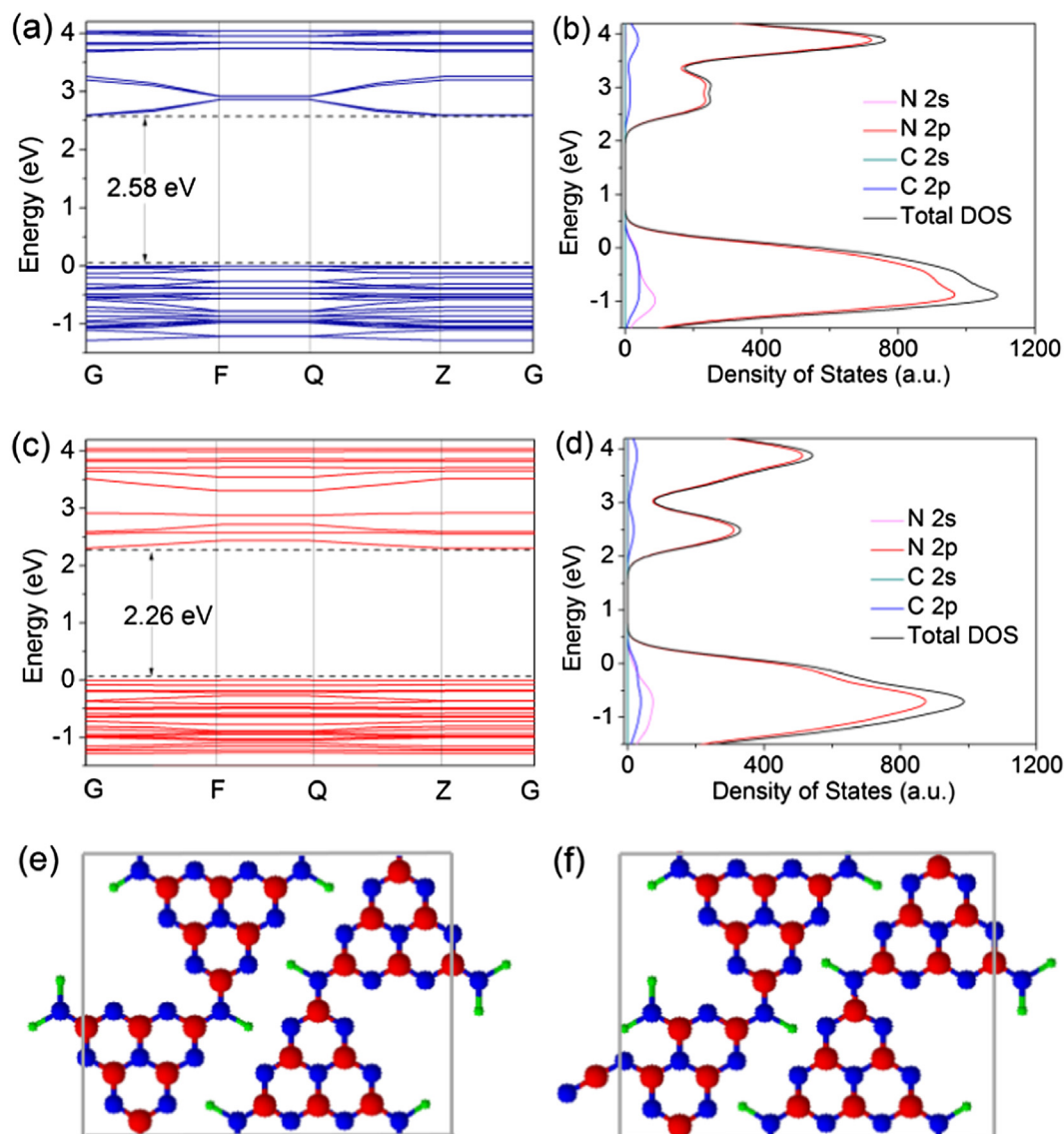


Fig. 3. (a) and (c) Calculated band structures. (b) and (d) Density of states. (e) and (f) Structure models of CNNS and DCNNS-550.

Table 1

Photocatalytic performance of various catalysts in visible light-driven oxidative coupling of benzylamines.^a

<chem>c1ccccc1CN</chem> $\xrightarrow[\text{CH}_3\text{CN}, 25^\circ\text{C}]{\text{Catalyst, visible light}}$ <chem>c1ccccc1C=Nc2ccccc2</chem>			
Entry	Catalyst	Con. (%)	Sel. (%)
1	CNNS	26	99
2	DCNNS-450	35	99
3	DCNNS-500	46	99
4	DCNNS-550	53	99
5	DCNNS-600	44	99
6	BCN	7	99
7	DBCN-550	30	99
8 ^b	DCNNS-550	15	99
9 ^c	DCNNS-550	—	—
10 ^d	—	—	—

^a Reaction conditions unless noted otherwise: benzylamine (0.1 mmol), catalyst (10 mg), solvent (CH₃CN, 5 mL), O₂ (1 atm), light ($\lambda > 420$ nm), temperature (25 °C), time (2 h).

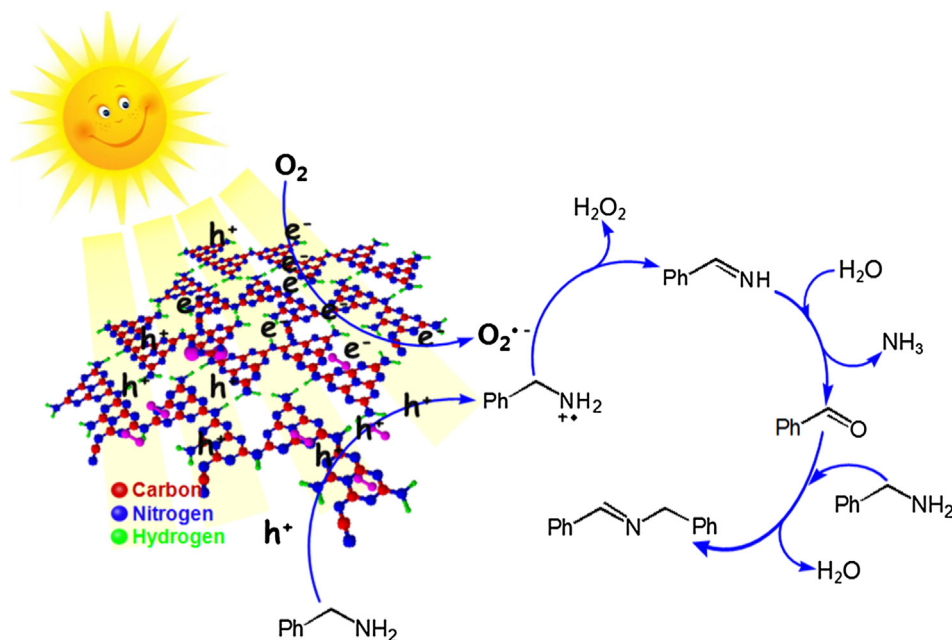
^b Under N₂ atmosphere.

^c Without light.

^d Without using catalyst.

shows the comparable catalytic performance (71% conversion for 2 h and 91% conversion for 8 h, Fig. S9). However, as is said from the reference, even more ammonium sulfate added led to a complete decomposition of carbon nitride and thus rather low yields of solid materials. Therefore, by the referenced method for the m-O=C₃N₄-2.3, the very large mass loss of the g-C₃N₄ would be taken place. If the ammonium sulfate was decreased to prepare the m-O=C₃N₄-2.3, and only 59% of conversion was obtained. However, in this work, the DCCNS-550 was prepared by the thermal post-treatment of CNNS that formed at 550 °C in the N₂ atmosphere. The mass loss of sample is very low, suggesting the great potential for practical applications.

Experimental studies and theoretical calculation reveal that the homojunction-like structure is constructed in DCNNS owing to the introduction of cyano-groups, which can facilitate the generation of hot carriers, and thus promotes the O₂^{•−} production. Based on the aforementioned results and references [10], we proposed the plausible reaction mechanism (Scheme 2). At first, O₂ molecules were activated by photoinduced electrons to form O₂^{•−} species over cyano groups decorated g-C₃N₄ nanosheets, while the benzylamine loses an electron by holes to form benzylammonium cation



Scheme 2. Plausible mechanism of visible light photocatalytic oxidative coupling of benzylamine to its corresponding imine over the cyano groups decorated g-C₃N₄ nanosheets.

Table 2

Visible light-driven oxidative coupling of diverse amines to their corresponding imines over the optimized cyano-decorated carbon nitride nanosheets (DCNNS-550) catalyst.^a

Entry	Substrate	Product	Con. (%)	Sel. (%)
1			99	99
2			99	96
3			99	99
4			96	97
5			99	96
6			99	98
7			94	97
8			97	90
9			99	99

^a Substrate (0.1 mmol), catalyst (25 mg), acetonitrile (5 mL), O₂ (1 atm), light source: $\lambda > 420$ nm, time (4 h), temperature (25 °C).

radical. The as-formed benzylammonium cation radical is oxidized by the O₂^{•−} radical to produce phenylacetaldehyde and NH₃. Finally, the phenylacetaldehyde was suffered from the condensation process with another benzylamine to form the corresponding imine.

Furthermore, the promoting effect of cyano groups those anchored on carbon nitride nanosheets on the visible light-driven molecular oxygen activation was further examined by changing diverse substituted benzylamines as substrates. The reaction tests were performed on the optimized DCNNS-550, and the results are listed in Table 2. The substituted benzylamines bearing both electron-rich groups (CH₃ and OCH₃) and electron-deficient groups (F, Cl and Br) were converted into their corresponding imines with

excellent conversion and high selectivity. Moreover, the stability and recyclability of DCNNS-550 are also investigated, which are crucial metrics for practical applications. From Figs. S10 and S11, the catalytic activity of DCNNS-550 shows a little decrease within four-cycle test, and well retains its $\text{—C}\equiv\text{N}$ groups after the oxidative process, clearly demonstrating it good recyclability and reusability.

4. Conclusions

We demonstrated that the introduced cyano groups in g-C₃N₄ nanosheets by a facile and green re-heating approach can efficiently promote visible light-driven molecular oxygen activation.

Experimental characterizations and DFT calculations reveal that the formation of homojunction-like structure in DCNNS can efficiently facilitate the generation of hot carriers, thus, promotes the $O_2^{\cdot-}$ production. The DNNS-550 shows 2.4 times higher activity than the pristine CNNS for the oxidative coupling of benzylamine to imine. The cyano group-promoting strategy has been extended to diverse amines to their corresponding imines. It is envisaged that this discover also paves a new way to design novel and efficient g-C₃N₄-based materials for photocatalytic and the other applications.

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References

- [1] J. Kou, C. Lu, J. Wang, Y. Chen, Z. Xu, R.S. Varma, *Chem. Rev.* 117 (2017) 1445–1514.
- [2] X. Lang, H. Ji, C. Chen, W. Ma, J. Zhao, *Angew. Chem. Int. Ed.* 50 (2011) 3934–3937.
- [3] N. Zhang, X. Li, H. Ye, S. Chen, H. Ju, D. Liu, Y. Lin, W. Ye, C. Wang, Q. Xu, J. Zhu, L. Song, J. Jiang, Y. Xiong, *J. Am. Chem. Soc.* 138 (2016) 8928–8935.
- [4] F. Raza, J.H. Park, H.R. Lee, H.I. Kim, S.J. Jeon, J.H. Kim, *ACS Catal.* 6 (2016) 2754–2759.
- [5] S. Sarina, H. Zhu, E. Jaatinen, Q. Xiao, H. Liu, J. Jia, C. Chen, J. Zhao, *J. Am. Chem. Soc.* 135 (2013) 5793–5801.
- [6] N. Zhang, Y. Zhang, M.Q. Yang, Z.R. Tang, Y.J. Xu, *J. Catal.* 299 (2013) 210–221.
- [7] N. Zhang, Y. Zhang, M.Q. Yang, Y.J. Xu, *Curr. Organ. Chem.* 17 (2013) 2503–2515.
- [8] N. Zhang, M.Q. Yang, S. Liu, Y. Sun, Y.J. Xu, *Chem. Rev.* 115 (2015) 10307–10377.
- [9] M.Q. Yang, Y.J. Xu, *Phys. Chem. Chem. Phys.* 15 (2013) 19102–19118.
- [10] F. Su, S.C. Mathew, L. Moehlmann, M. Antonietti, X. Wang, S. Blechert, *Angew. Chem. Int. Ed.* 50 (2011) 657–660.
- [11] Y. Zhao, M. Antonietti, *Angew. Chem. Int. Ed.* 56 (2017) 9336–9340.
- [12] M.B. Khaled, R.K. El Mokadem, J.D. Weaver III, *J. Am. Chem. Soc.* 139 (2017) 13092–13101.
- [13] F. Raza, D. Yim, J.H. Park, H.I. Kim, S.J. Jeon, J.H. Kim, *J. Am. Chem. Soc.* 139 (2017) 14767–14774.
- [14] X.C. Wang, K. Maeda, A. Thomas, K. Takanabe, G. Xin, J.M. Carlsson, K. Domen, M. Antonietti, *Nat. Mater.* 8 (2009) 76–80.
- [15] W.J. Ong, L.L. Tan, Y.H. Ng, S.T. Yong, S.P. Chai, *Chem. Rev.* 116 (2016) 7159–7329.
- [16] D. Zhang, Y. Guo, Z. Zhao, *Appl. Catal. B: Environ.* 226 (2018) 1–9.
- [17] P. Yang, H. Ou, Y. Fang, X. Wang, *Angew. Chem. Int. Ed.* 56 (2017) 3992–3996.
- [18] P. Xia, B. Zhu, J. Yu, S. Cao, M. Jaroniec, *J. Mater. Chem. A* 5 (2017) 3230–3238.
- [19] L. Shi, K. Chang, H. Zhang, X. Hai, L. Yang, T. Wang, J. Ye, *Small* 12 (2016) 4431–4439.
- [20] Y. Li, S. Ouyang, H. Xu, X. Wang, Y. Bi, Y. Zhang, J. Ye, *J. Am. Chem. Soc.* 138 (2016) 13289–13297.
- [21] Z. Zhao, G. Ge, D. Zhang, *ChemCatChem* 10 (2018) 62–123.
- [22] S. Verma, R.B.N. Baig, M.N. Nadagouda, R.S. Varma, *ACS Sustain. Chem. Eng.* 5 (2017) 3637–3640.
- [23] S. Verma, R.B.N. Baig, M.N. Nadagouda, R.S. Varma, *ACS Sustain. Chem. Eng.* 4 (2016) 2333–2336.
- [24] X.H. Li, X.C. Wang, M. Antonietti, *Chem. Sci.* 3 (2012) 2170–2174.
- [25] Y. Zhao, M. Shalom, M. Antonietti, *Appl. Catal. B: Environ.* 207 (2017) 311–315.
- [26] R.B.N. Baig, S. Verma, M.N. Nadagouda, R.S. Varma, *Green Chem.* 18 (2016) 1019–1022.
- [27] P. Sharma, Y. Sasson, *Green Chem.* 19 (2017) 844–852.
- [28] H. Wang, S.L. Jiang, S.C. Chen, D.D. Li, X.D. Zhang, W. Shao, X.S. Sun, J.F. Xie, Z. Zhao, Q. Zhang, Y.P. Tian, Y. Xie, *Adv. Mater.* 28 (2016) 6940–6945.
- [29] H. Wang, X. Sun, D. Li, X. Zhang, S. Chen, W. Shao, Y. Tian, Y. Xie, *J. Am. Chem. Soc.* 139 (2017) 2468–2473.
- [30] J.J. Zhang, J.M. Ge, H.H. Wang, X. Wei, X.H. Li, J.S. Chen, *ChemCatChem* 8 (2016) 3441–3445.
- [31] A. Kumar, P. Kumar, C. Joshi, S. Ponnada, A.K. Pathak, A. Ali, B. Sreedhar, S.L. Jain, *Green Chem.* 18 (2016) 2514–2521.
- [32] S. Samanta, S. Khilari, D. Pradhan, R. Srivastava, *ACS Sustain. Chem. Eng.* 5 (2017) 2562–2577.
- [33] S. Samanta, R. Srivastava, *Appl. Catal. B: Environ.* 218, 621–636; (c) X.H. Li, M. Antonietti, *Chem. Soc. Rev.* 42 (2013) 6593–6604.
- [34] H. Yu, R. Shi, Y. Zhao, T. Bian, Y. Zhao, C. Zhou, G.I.N. Waterhouse, L.Z. Wu, C.H. Tung, T. Zhang, *Adv. Mater.* 29 (2017) 1605148.
- [35] G. Liu, G. Zhao, W. Zhou, Y. Liu, H. Pang, H. Zhang, D. Hao, X. Meng, P. Li, T. Kako, J. Ye, *Adv. Funct. Mater.* 26 (2016) 6822–6829.
- [36] V.W. Lau, I. Moudrakovski, T. Botari, S. Weinberger, M.B. Mesch, V. Duppel, J. Senker, V. Blum, B.V. Lotsch, *Nat. Commun.* 7 (2016) 12165.
- [37] L. Lina, C. Daniel, S. Aleksandr, G. Zhang, M. Antonietti, Y. Zhao, *Appl. Catal. B: Environ.* 229 (2017) 249–253.
- [38] P. Niu, M. Qiao, Y. Li, L. Huang, T. Zhai, *Nano Energy* 44 (2018) 73–81.
- [39] X. Lu, K. Xu, P. Chen, K. Jia, S. Liu, C. Wu, *J. Mater. Chem. A* 2 (2014) 18924–18928.
- [40] H. Wang, X. Zhang, J. Xie, J. Zhang, P. Ma, B. Pan, Y. Xie, *Nanoscale* 7 (2015) 5152–5156.
- [41] N. Zhang, X. Li, Y. Liu, R. Long, M. Li, S. Chen, Z. Qi, C. Wang, L. Song, J. Jiang, Y. Xiong, *Small* 13 (2017) 1701354.